



# Exploring the status of artificial intelligence for healthcare research in Africa: a bibliometric and thematic analysis

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## Abstract

This paper explores the status of Artificial Intelligence (AI) for healthcare research in Africa. The aim was to use bibliometric and thematic analysis methods to determine the publication counts, leading authors, top journals and publishers, most active institutions and countries, most cited institutions, funding bodies, top subject areas, co-occurrence of keywords and co-authorship. Bibliographic data were collected on April 9 2022, through the Lens database, based on the critical areas of authorship studies, such as authorship pattern, number of authors, etc. The findings showed that several channels were used to disseminate the publications, including articles, conference papers, reviews, and others. Publications on computer science topped the list of documented subject categories. The Annals of Tropical Medicine and Public Health is the top journal, where articles on AI have been published. One of the top nations that published AI research was the United Kingdom. With 143 publications, Harvard University was the higher education institution that produced the most in terms of affiliation. It was discovered that the Medical Research Council was one of the funding organizations that supported research, resulting in the publication of articles in AI. By summarizing the current research themes and trends, this work serves as a valuable resource for researchers, practitioners, and funding organizations interested in Artificial intelligence for healthcare research in Africa.

**Keywords** Artificial Intelligence · Machine learning · Deep learning · Healthcare · Bibliometric analysis · Thematic analysis

## 1 Introduction

A healthcare revolution is currently unfolding, driven by escalating total healthcare expenditures and a shortage of physicians. In response to these challenges, various sectors are adopting state-of-the-art Information and Communication Technology (ICT)-based processes and solutions. These innovations hold the potential to reduce costs and provide effective solutions for the evolving healthcare landscape. Technology is evolving toward medical solutions

that “provide intelligent solutions for evidence and outcome-based health, focused on collaborative and preventative care”, according to Frost and Sullivan [1]. Robotics, virtual and augmented reality and artificial intelligence (AI) can all be used to create these intelligent solutions.

Artificial intelligence (AI) was first developed in the 1950s, and as science and technology have advanced, its use in modern healthcare has exploded [2]. AI is beginning to impact almost every aspect of healthcare, including clinical decision support at points of care and patient self-management of chronic conditions at home. Implementing AI technology in the healthcare sector promotes disease prediction, diagnosis, and treatment, which is advantageous to both patients and healthcare professionals [3]. By quickly evaluating patients’ electronic health records on both the horizontal and vertical axes, AI can emulate the diagnostic power of human doctors to increase the accuracy of diagnoses [4]. In general, AI systems are often driven by data streams and cutting-edge algorithms, which enhance healthcare by preventing hospitalizations, lowering complications, lowering administrative costs, and raising patient participation. The

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continuing shift from our current disease-based system to one focused on prevention and health management promises to be accelerated and scaled by AI systems [5].

Notably, due to massive availability of data in healthcare sectors across different countries, it has been easy to develop AI solution in healthcare domain. This has resulted to a widespread use of AI in medicine [6], particularly to aid in illness identification and prevention [7–10]. In particular, specialized care (such as radiology and pathology) is experiencing change as a result of AI [11–13]. Large data sets are becoming more widely available, and new analytical techniques that rely on them have encouraged this development. As a result, Global recognition has emerged in recent years of the significance of AI-based new technologies to advance healthcare, shifting away from hospital-cantered systems toward integrated care, boosting health promotion and illness prevention, and introducing customized medicine.

In addition, a variety of AI and Machine Learning (ML) approaches have been applied to evaluate the vast amounts of data in the healthcare industry. For instance, a prediction model based on logistic regression was used to automate the early diagnosis of cardiac disease with promising results [14]. Medical imaging has also made the use of ML to automatically identify object features realistic [15]. Moreover, AI technologies could assist doctors provide better patient care and lead to more precise diagnosis of medical issues, such as surgery.

In spite of the fact that AI has made remarkable progress across numerous fields, yet it faces notable challenges in delivering precise predictions in certain areas. A prime example of such a domain is mental health and psychiatry, where conditions are exceptionally intricate. The outcomes in this realm are shaped by a myriad of factors, encompassing genetics, the surrounding environment, and individual life experiences [16]. Furthermore, mental health diagnoses often rely on subjective criteria, making it challenging for AI models to predict treatment responses or long-term outcomes effectively [17]. Developing AI models for detecting mental health and psychiatry typically require large and diverse data sets to make accurate predictions. However, with rare diseases, there may not be enough data available to train models effectively. In addition, rare diseases are highly heterogeneous, meaning that each case is unique, making it difficult for AI to provide precise predictions for individuals with these conditions [18].

Another complex area in which AI has limitations is the decision-making process within the legal and judicial systems. Although artificial intelligence can help with legal research and document analysis, determining the results of legal disputes is still a difficult endeavour [19]. Complex rules, ever-evolving case law, and nuanced human judgment are necessary components of legal systems. To accurately

predict legal outcomes, one must have a comprehensive knowledge of legal precedent and the ability to take into account human subjectivity as well as ethical considerations [20].

Moreover, AI's effectiveness in predicting outcomes related to socioeconomic factors, such as an individual's future income or job prospects, remains limited. These predictions of socioeconomic factors using AI depend on a myriad of variables, including personal choices, economic conditions, and societal influences. While AI can analyse historical data and identify correlations, it often struggles to capture the complexity of human decision-making and the dynamic nature of socioeconomic factors [21].

In these multifaceted domains, human expertise, ethical considerations, and a cautious approach to AI integration are essential. It is essential to recognize that the “implementation gap” in the adoption of AI for routine clinical care is a pervasive obstacle, and that its potential impact extends to African nations as well. While industrialized nations have made significant advances in AI research and development for healthcare, the complexity and limited resources of the African healthcare landscape may exacerbate this problem. One major concern is the digital divide in healthcare infrastructure [22]. Many regions in Africa still lack the necessary digital infrastructure, including reliable internet connectivity and electronic health records systems, which are fundamental for AI integration. This digital divide can hinder the collection, storage, and sharing of healthcare data, a crucial component of AI-driven applications [23]. Another significant challenge lies in the shortage of healthcare professionals and AI experts [24]. Implementing AI in clinical settings requires a skilled workforce capable of developing, maintaining, and effectively using AI systems. African countries often face shortages of healthcare workers, let alone AI specialists, which can impede the adoption of AI-driven solutions [25].

In Africa, AI is beginning to fill the staffing and resource gaps that commonly exist in healthcare institutions. Olk [26] contend that AI in African healthcare has the potential to significantly improve care on the continent, especially for marginalized and vulnerable people. However, the majority of AI advancements in healthcare are geared toward meeting the demands of industrialized nations, where the majority of research is done. With this in mind, a study to investigate the status of Artificial Intelligence (AI) for healthcare research in Africa is of paramount importance. Therefore, the main objective of this study is to examine the publishing trends of artificial intelligence for healthcare in Africa. The aim is to ascertain the research endeavours pertaining to artificial intelligence (AI) within the healthcare domain, specifically focusing on studies undertaken in the African continent. Hence, this study conducted a bibliometric and thematic analysis of

publishing trends of Artificial intelligence for healthcare in Africa by answering the following research questions:

Research question 1 (RQ1): What is the current trend of publications in Artificial intelligence for healthcare in Africa?

Research question 2 (RQ2): Which are the most productive and influential countries, institutions and authors on Artificial intelligence for healthcare in Africa?

Research question 3 (RQ3): Which are the most prevalent themes of Artificial intelligence for healthcare in Africa?

The rest of this study is organized as follows: section two illustrates the literature reviews, including hospital information system, and related works. The third section reports the study's methods and data, including methods, data sources and data analysis. Section four illustrates the results of the study, while section five reports the discussions of the study, including characteristics of publications, influential countries/regions and institutions, and science mapping analysis. The last section ends this paper with the conclusions.

## 2 Literature reviews

### 2.1 Hospital Information System (HIS)

HIS is a computer system created to manage all of the hospital's medical and administrative data, so that medical staff may carry out their duties more successfully and effectively, according to Biomedical Informatics [27]. In addition, HIS oversees all hospital information processing activities to ensure high-quality patient care and medical research [28]. Clinical Information System (CIS), Financial Information System (FIS), Laboratory Information System (LIS), Nursing Information System (NIS), Pharmacy Information System (PIS), Picture Archiving and Communication System (PACS), and Radiology Information System (RIS) are the minimum components of HIS. Each category has a specific function, department, and user base to improve hospital services.

A lot of data can be stored in the HIS. The amount of data that may be collected by a clinical institution and used for research and AI applications is unfathomable. Descriptive and epidemiological research could effectively undergo a revolution if millions of data points from a single topic were made available. In addition, it can prepare ML, neural network (NN), or deep learning (DL) algorithms for clinical situation prediction and support clinical decision-making for doctors.

### 2.2 Related works

Bibliometrics has become crucial for analysing and predicting research trends in recent years [29]. A research stream can be assessed using bibliometric techniques that can add objectivity and reduce researcher bias, as indicated by [30]. As a result, researchers are becoming more interested in bibliometric methods as a trustworthy and impersonal way of research analysis [31, 32].

The reported scientific articles reveal significant variations in terms and previously researched research areas. Hao et al. [33]'s bibliometric analysis focuses on text mining in medical studies. Text mining is the application of data mining techniques to text data. Text mining automatically unearths fresh, previously undiscovered data using a computer to extract data from various text sources. Similar to this, dos Santos et al. [34] studies use machine learning (ML) and data mining tools to address public health issues. According to the findings of this study, public health can be summed up as the practice of disease prevention, wellness promotion, and life extension. Finding new information that would otherwise be concealed is made feasible by data mining and machine learning approaches.

More recently, Choudhury et al. [35] give a systematic review on the application of machine learning (ML) to enhance the care of older people, highlighting eligible studies, mostly in the areas of psychological problems and visual ailments. Tran et al. [36] focus on the international development of AI research in medicine. Their bibliometric study identifies trends and subjects about AI techniques and applications.

The field of AI research is also expanding in a new direction. In addition, Guo et al. [37]'s bibliometric analysis offers a thorough examination of AI papers until December 2019. The report focuses on practical AI health applications and provides academics with an understanding of how algorithms might benefit medical professionals. Choudhury and Asan's [38] scientific contribution offers a thorough analysis of the AI literature to detect patient health hazards. Their study covers 53 studies involving technology for clinical alerts, clinical reports, and drug safety. For several reasons, our analysis is different from the existing literature. For academics and practitioners using the lens database, it seeks to give a bibliometric and thematic analysis of characteristics including authors, nations, citations, and keywords to help shape future research ideas. Furthermore, we did not confine our investigation to a particular time frame.

### 3 Methods and data

#### 3.1 Methods

Bibliometrics's main goal is finding quantitative information on written materials, such as output, influence, and collaboration [39]. The bibliometric analysis and visualization of AI research in healthcare served as the foundation for this study. Our method has successfully captured the temporal content of research topics and spotted research trends, which is what we want to do in this work [40]. The following four procedures from [41] were employed in the bibliometric analysis: (i) identify and gather relevant literature on the subject; (ii) carry out bibliometric analysis; (iii) present and discuss the findings; and (iv) describe the academic field and the context of the study.

#### 3.2 Data sources

On April 9, 2022, the researchers searched the Lens database for publications that had been published in the subject of AI for healthcare. This search encompassed all varieties of research articles and study designs. The search string ("Artificial Intelligence" OR "Machine Learning" OR "Deep Learning" OR "Natural Language Processing") AND "Health\*" AND "Africa") was used, and no constraint was set on the year of publication. It is important to note that Africa was included in the search string to ensure that the retrieved results from the lens database focus on the African continent. A similar search was conducted by [42] and Phoobane et al. [43]. Furthermore, the inclusion and exclusion criteria used in this study are shown in Table 1.

#### 3.3 Data analysis

VOSviewer allowed us to deepen the investigation on co-occurrence and co-authors relationships. Bibliometrix was additionally used to create a three-field plot, topic trends, and thematic map [44]. Concerning descriptive data analysis, Microsoft Excel™ was used to elaborate statistics about the evolution over time of the topics, the citations, journal releasing, authorship pattern, publication source, leading authors, journals, institutions, nations and references, the keywords, the authors and the geographical distribution of

the studies. This paper's conclusion has been drawn based on the analysis, which has fulfilled the study's objective.

### 4 Results

#### 4.1 Publication counts per year

Results in Fig. 1 show that 5,309 scholarly publications have been published as of various document kinds in the Lens database. Figure 1 also shows that there was no appreciable increase in the number of publications on AI for healthcare in Africa between 1960 and 2000. Researchers may have engaged in research activities, but their scholarly output was not available online. After 2000, there was a change, as depicted in Fig. 1. As a result, since 2000, scholars have been publishing in the topic, and their academic writing is readily available online.

#### 4.2 Avenue through which publications were communicated

The results obtained in this study indicate that 4270 research articles were published in journals, followed by book chapters (237), conference proceedings articles (193), books (134), preprint (78), unknown (56), and reports (41). It is important to note that the avenue through which most of the publications were communicated is the journals, where 4270 (80.43%) research articles were published. The rest of the publication avenues, such as conference proceedings, journal issue, component, data set, dissertation, editorial, news, other, and reference entry, were minimal (i.e., below 40), as shown in Fig. 2.

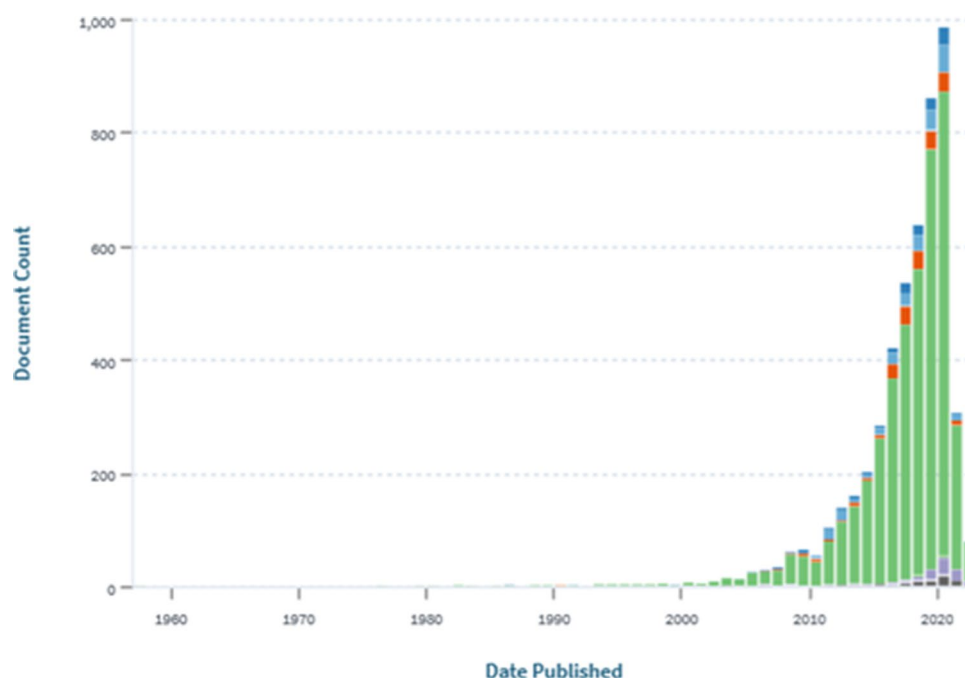
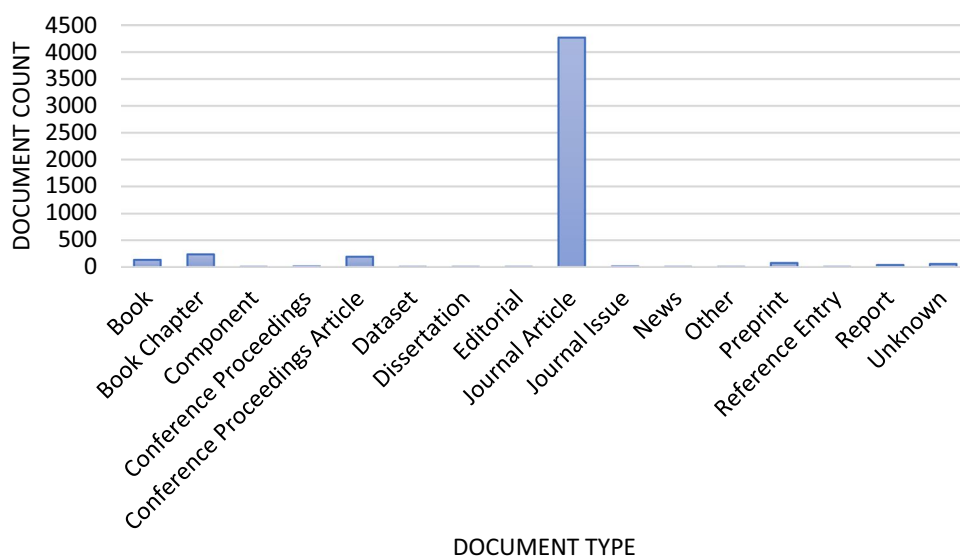
#### 4.3 Top subject areas

The distribution of publications across various fields of study, as shown in Fig. 3, reflects the interdisciplinary nature and wide-ranging applications of AI and ML in healthcare research. The significance of these top fields is described as follows:

Computer Science (1070 publications): AI and ML are deeply rooted in computer science [45]. Research in this field often focuses on algorithm development, data processing, and computational techniques that underpin AI applications in healthcare.

**Table 1** Inclusion and exclusion criteria

| Inclusion criteria                                            | Exclusion criteria                    |
|---------------------------------------------------------------|---------------------------------------|
| Papers published since the arrival of AI until 9th April 2022 | Papers published from 10th April 2022 |
| Papers published in the Lens database                         | Papers published in other databases   |
| Papers written in English language                            | Non-English language articles         |

**Fig. 1** Publication counts per year**Fig. 2** Avenue through which publications were communicated

**Medicine (881 publications):** Medical professionals and researchers are increasingly leveraging AI to enhance diagnostics, treatment planning, and patient care [46]. AI has the potential to revolutionize medical practice.

**Biology (524 publications):** AI is used in bioinformatics and genomics to analyse biological data, model complex biological systems, and understand genetic variations and disease mechanisms [47].

**Artificial Intelligence (510 publications):** This category likely includes research that is directly related to AI methods and their applications in healthcare, such as natural language processing, computer vision, and predictive modelling [48].

**Psychology (481 publications):** AI can be applied to behavioural analysis, mental health diagnostics, and personalized interventions, making it relevant to psychology and mental healthcare [49].

**Population (338 publications):** Population health research can benefit from AI-driven insights into disease trends, healthcare disparities, and interventions to improve public health outcomes [50].

**Business (314 publications):** AI has economic implications for healthcare systems and industries. Research may cover the business aspects of AI adoption, cost-effectiveness, and market trends [51].



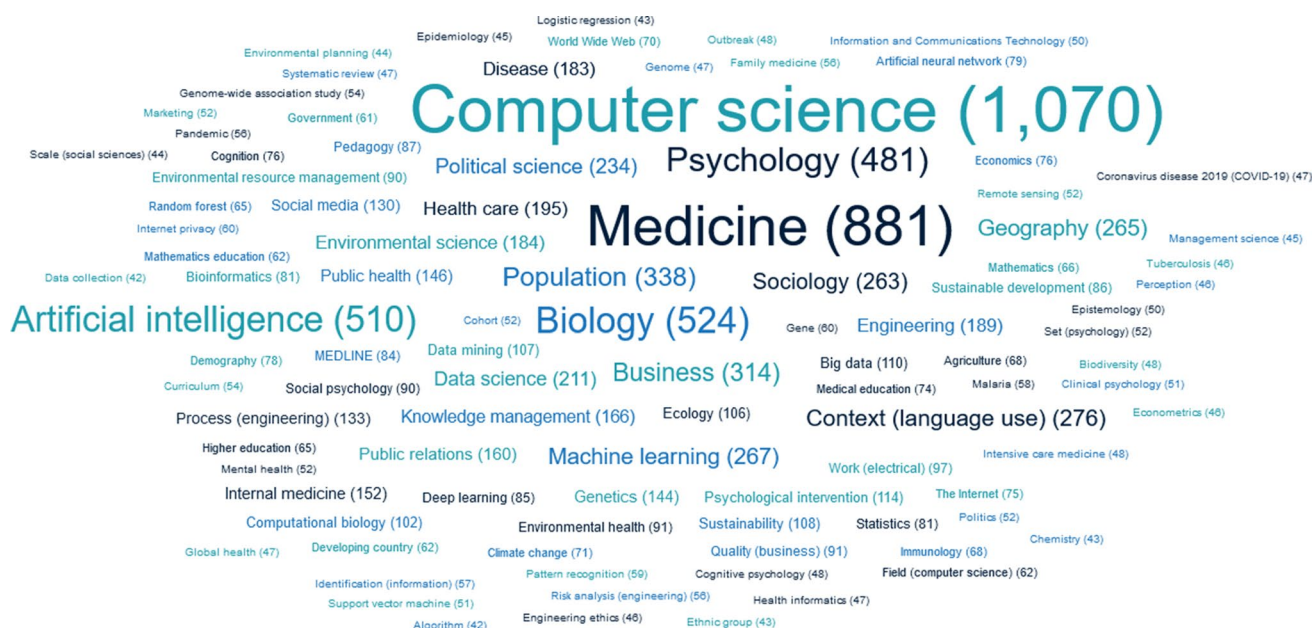


Fig. 3 Top fields of study

Context (Language Use) (276 publications): NLP techniques are essential in healthcare for extracting meaningful information from clinical notes, patient records, and medical literature [52].

Machine learning (267 publications): ML, a subset of AI, plays a fundamental role in healthcare data analysis, predictive modelling, and decision support systems [53].

Geography (265 publications): Geographic information systems (GIS) and spatial analysis, often combined with AI, can aid in healthcare planning, disease mapping, and resource allocation [54].

This broad distribution highlights AI's capacity to influence various facets of healthcare, encompassing clinical practice, medical research, public health, and business operations [55]. Collaborative efforts spanning these domains are imperative to fully leverage AI's potential in healthcare and tackle the multifaceted challenges within this sphere.

#### 4.4 Top journals by publisher

Results in Fig. 4 indicate that the journal of *Annals of Tropical Medicine and Public Health*, which Africa Health Research Organization is publishing, is at the top with 220 publications, followed by *PLoS*, published by the Public Library of Science with 123 publications. The third journal is *Remote Sensing* (49 publications) which is published by MDPI AG, followed by *SSRN Electronic Journal* (39 publications), *Procedia Manufacturing* (36 publications), *Sustainability* (30 publications), *PLoS neglected tropical diseases* (29 publications), *PLoS computational biology*

(12 publications), *PLoS genetics* (12 publications), *BMC genomics* (10 publications), *Expert Systems with Applications* (10 publications), and *Energies* (10 publications).

#### 4.5 Top publisher in the world

Figure 5 indicates that Elsevier is the first with 620 publications. Africa Health Research Organization is the second with 224 publications. Springer Nature is the third with 218 publications, followed by Wiley with 210 publications, Public Library of Science with 193 publications, MDPI AG (171 publications), BioMed Central (152 publications), Nature Publishing Group (121 publications), Informa UK Limited (120 publications), and International Institute for Science, Technology, and Education (109 publications). These are the ten top publishers of AI research in the World.

#### 4.6 Most active countries in the world

As shown in Fig. 6, the findings identified the countries in the World, where publications in AI for healthcare are published. The United Kingdom (1391 publications) was the first with the largest number of publications in AI, followed by the United States (1343 publications), South Africa (350 publications), Australia (291 publications), Germany (260 publications), Netherlands (227 publications), Spain (208 publications), Canada (198 publications), Switzerland (184 publications), and China (179 publications). It is important to note that South Africa is among the top three countries contributing to developing AI for Healthcare research in Africa. However, no other African country is included in

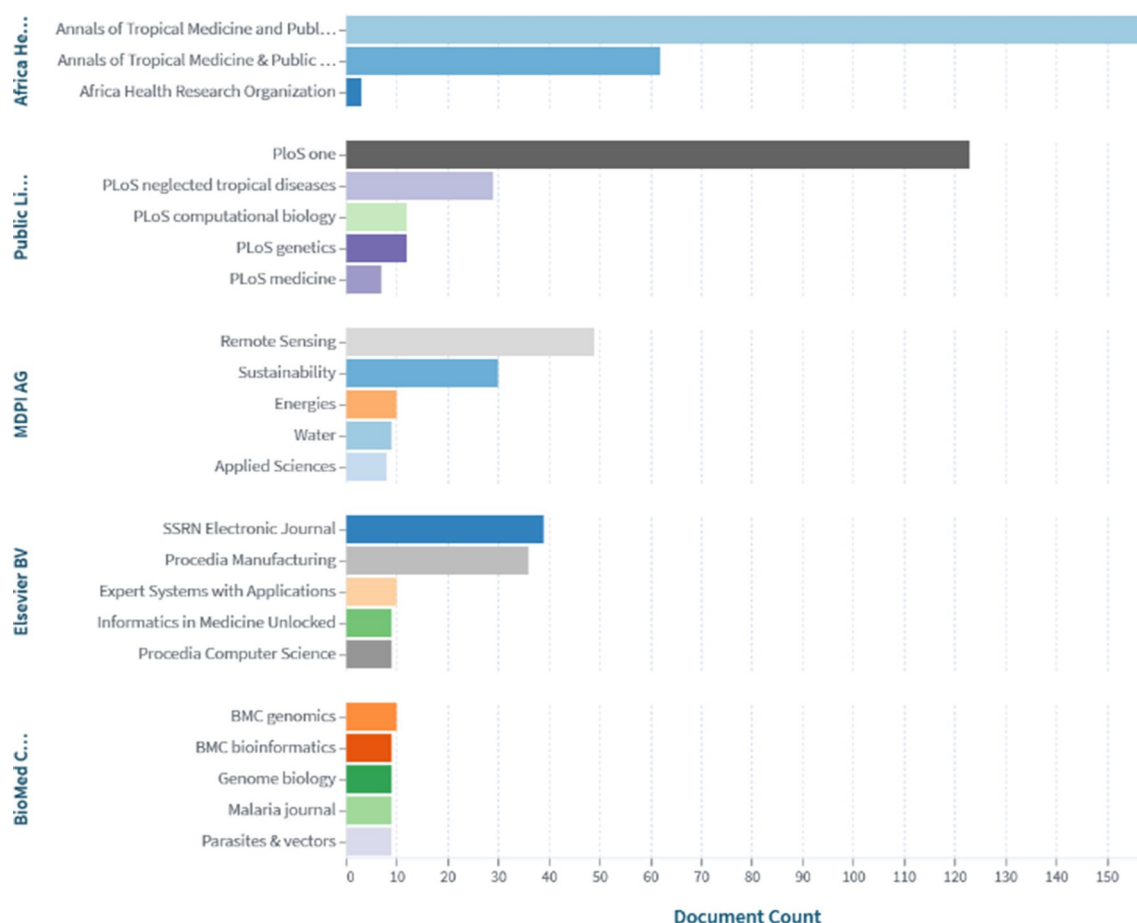


Fig. 4 Top journals by publisher

the list of the most active countries in the world. Therefore, other African countries need to pull up their socks to do well in AI for healthcare research in Africa.

#### 4.7 Most active countries in Africa

The top ten productive countries in Africa are summarised in Fig. 7. While analysing the leading countries, we found that South Africa, Nigeria, and Kenya are the most influential countries in the scientific production of AI for healthcare. From Fig. 7, we can also see that the number of published articles by South Africa is almost five times (350) compared to 2nd productive country, Nigeria (77) and seven times compared to the third productive country, Kenya (49). Other countries include Uganda (29 publications), Ghana (25 publications), Ethiopia (22 publications), Tanzania (17 publications), Egypt (10 publications), Mali (9 publications), Cameroon (8 publications), Botswana (8 publications), Malawi (8 publications), and Morocco (8 publications).

#### 4.8 Productivity by institution affiliation

Harvard University has the most significant number of publications, comprising 143 papers. At the second position is the University of London, with 124 publications. The University College London is in the third position with 116 publications. The University of Edinburgh then followed it with 107 publications, followed by the University of Oxford (97 publications), King's College London (92), University of Cambridge (83), Stanford University (76), Imperial College London (73), University of Cape Town (73), and Massachusetts Institute of Technology (65). It is important to note that the University of Cape Town is the only university from Africa in the top 10 institutions that are more productive in AI for healthcare.

Furthermore, in the top 10 institutes, seven universities from the United Kingdom (England), three universities from the United States, and one from South Africa, as indicated in Fig. 8 and Table 2. Although South Africa is in the third position of the most active countries in the world regarding AI for healthcare research in Africa, it has only one

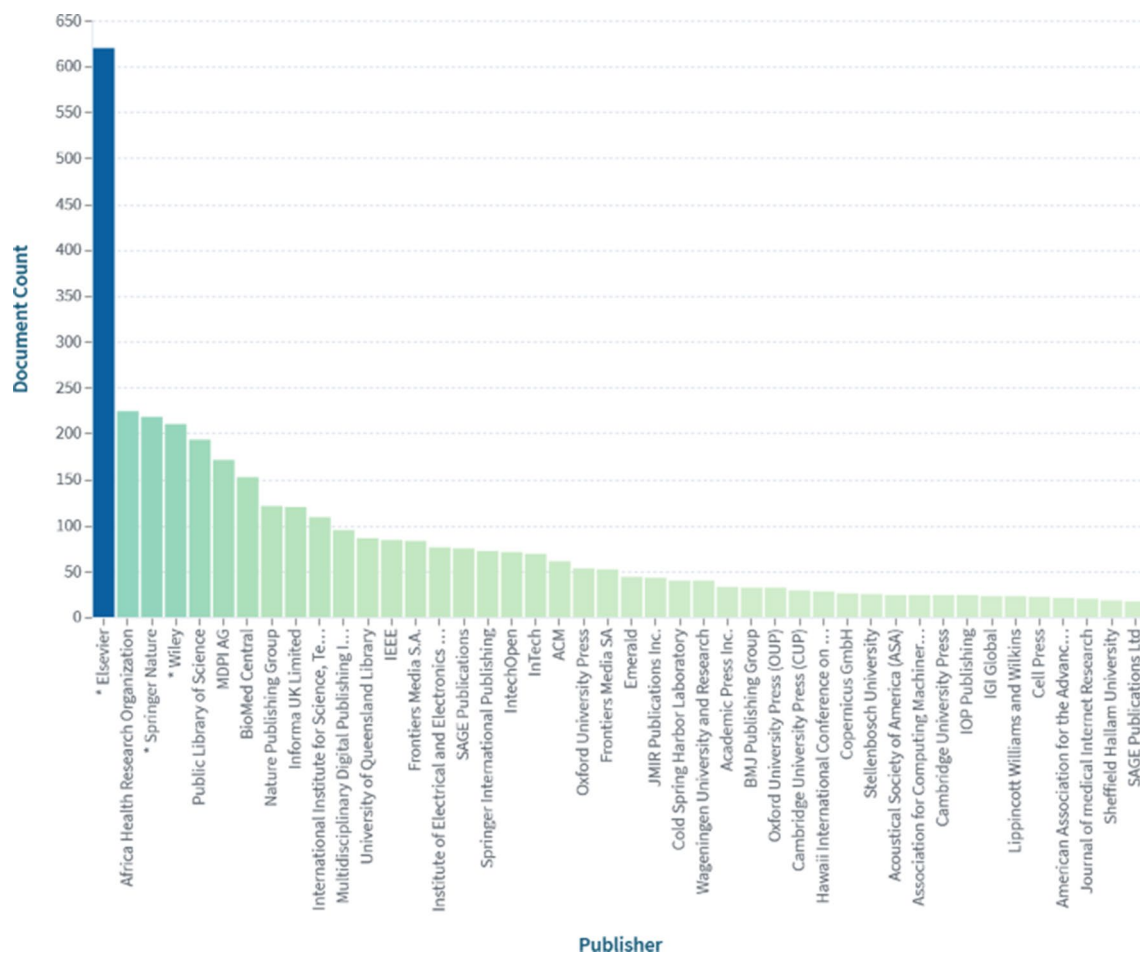
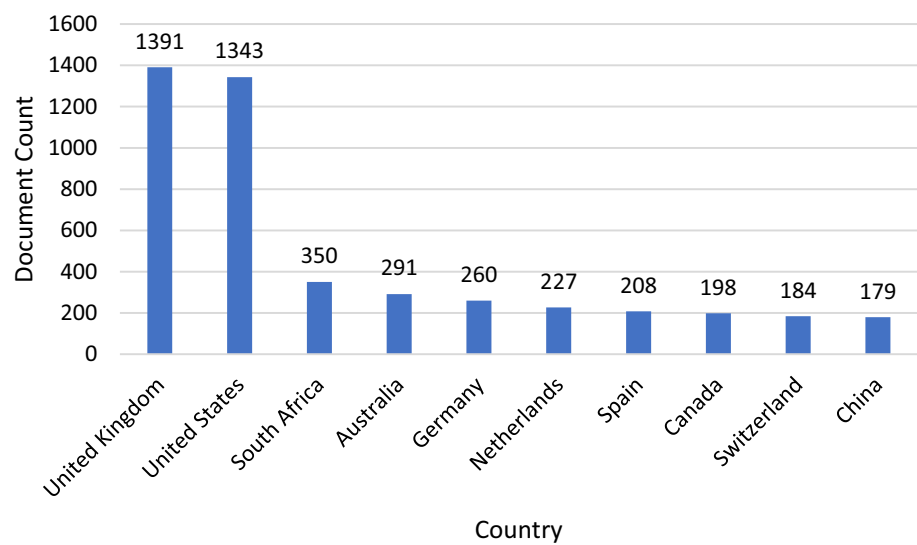
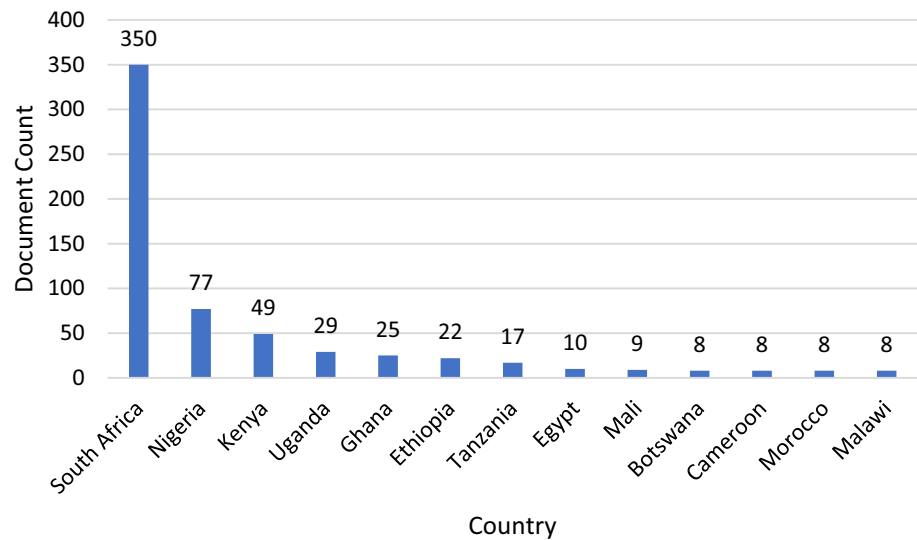
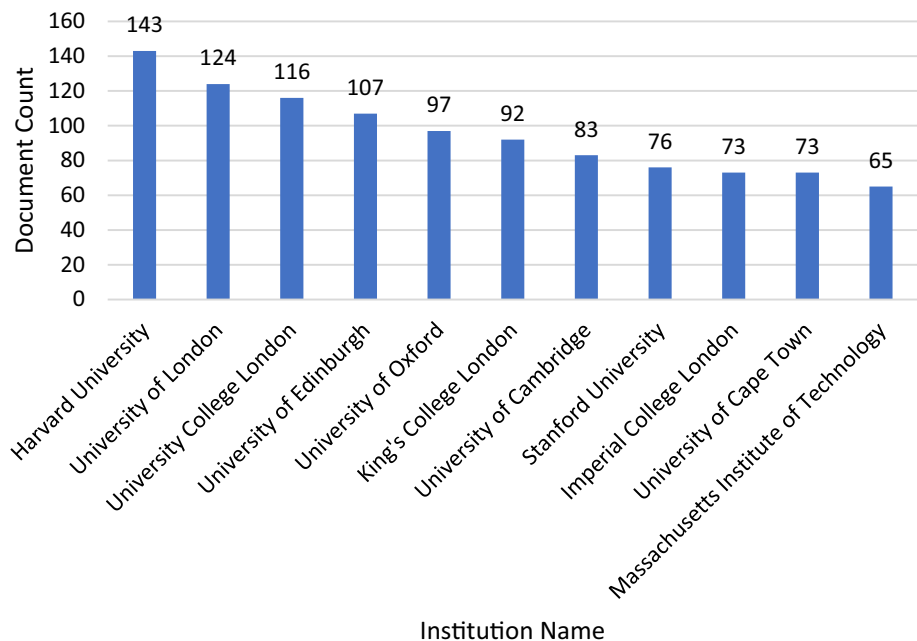


Fig. 5 Top publishers

Fig. 6 Most active countries in the world





**Fig. 7** Most active countries in Africa**Fig. 8** Top 10 institutions which are more productive in the area of AI for healthcare in Africa

University (the University of Cape Town), which is included in the list of the top institutions in the world. Therefore, more efforts are needed by South African institutions for them to be included in the leading institutions by document count in the world.

#### 4.9 Most cited institutions

Results in Table 3 indicate that Harvard University is at the top of the most cited institutions in the world concerning the research in AI for healthcare. The Broad Institute is the second. National Institutes of Health is the third, followed by the University of Michigan, Johns Hopkins University, University of Oxford, Massachusetts Institute of Technology,

Wellcome Trust Sanger Institute, Stanford University and Boston College. It should be noted that there is no African institution in the list of the top ten most-cited institutions. Hence, African institutions should make more efforts to publish research articles that will be cited frequently to improve the world's current status.

#### 4.10 Fields of study covered by the most active institutions

Regarding the areas of study covered by the most active institutions, Fig. 9 shows that King's College London is at the top in Medicine, while Harvard University is at the top in Biology. Furthermore, University College London is at the

**Table 2** Top institution name by document count

| Institution name                      | Country        | Document count |
|---------------------------------------|----------------|----------------|
| Harvard University                    | United States  | 143            |
| University of London                  | United Kingdom | 124            |
| University College London             | United Kingdom | 116            |
| University of Edinburgh               | United Kingdom | 107            |
| University of Oxford                  | United Kingdom | 97             |
| King's College London                 | United Kingdom | 92             |
| University of Cambridge               | United Kingdom | 83             |
| Stanford University                   | United States  | 76             |
| Imperial College London               | United Kingdom | 73             |
| University of Cape Town               | South Africa   | 73             |
| Massachusetts Institute of Technology | United States  | 65             |

**Table 3** Most cited institutions

| Institution name                      | Document count | Citing scholarly works |
|---------------------------------------|----------------|------------------------|
| Harvard University                    | 116            | 30,175                 |
| Broad Institute                       | 26             | 19,380                 |
| National Institutes of Health         | 26             | 12,326                 |
| University of Michigan                | 17             | 12,257                 |
| Johns Hopkins University              | 27             | 10,094                 |
| University of Oxford                  | 24             | 10,006                 |
| Massachusetts Institute of Technology | 40             | 8850                   |
| Wellcome Trust Sanger Institute       | 2              | 7704                   |
| Stanford University                   | 23             | 7544                   |
| Boston College                        | 2              | 7209                   |

top in Psychology. Cardiff University is at the top in Computer Science, and Harvard University is again at the top in Artificial Intelligence. Again, no African institution is at the top in any field of study. This implies that African countries should make more efforts to enable their institutions to compete with the rest of the world in all areas of study.

#### 4.11 Funding bodies

Regarding the funding bodies within and outside of Africa that facilitated the opportunity of publication for researchers in the areas of Artificial Intelligence for Healthcare in Africa, data in Fig. 10 shows that the medical research council is at the top of the list of organizations that sponsored the research activities. The Well come trust is the second. NIGMS NIH HHS is the third, followed by NIAID NIH HHS, NCI NIH HHS, NIMH NIH HHS, NHLBI NIH HHS, NHGRI NIH HHS, Biotechnology and Biological

Sciences Research Council and NCATS NIH HHS. The organizations specified are international organizations or research-related institutions supporting African research activities.

#### 4.12 Co-occurrence analysis of keywords

A co-occurrence analysis of 4267 keywords was conducted in this section. To show the keyword with the greatest total link strength (TLS), we set the slightest presence of a keyword at 10, and 30 keywords met the threshold. The method of normalization was association strength. Regarding clustering, the resolution was set at one and the least cluster size at one. Six clusters (Fig. 11) with 117 links and 220 total link strengths were identified. Clusters 1, 2, 3, 4, 5, and 6 consisted of 7, 7, 6, 5, 3, and 2 keywords, respectively, making a total of 30 keywords. Cluster 1 keywords include natural language processing, COVID-19, and health.

The number of keywords under each research domain, not their frequency, was used to build each cluster. The largest clusters are clusters 1 and 2, while cluster 6 is the smallest. Remember that the same cluster keywords are either those with close relationships or regularly appear together in published publications to forecast infectious diseases in Africa.

Among the ten most common author keywords shown in Table 4, it is essential to note that machine learning is the most frequent and most related keyword, followed by artificial intelligence, COVID-19, and epidemiology. It is worth noting that ML is in cluster 2, AI is in cluster 3, COVID-19 is in cluster 1, and epidemiology is in cluster 4.

Table 4 shows the total link strengths and the frequencies of the top keywords. The keyword “machine learning” has the highest frequency of 129. Other keywords with a high frequency include “artificial intelligence” (42), “epidemiology” (35), and “COVID-19” (33). It should be noted that the top 4 diseases are COVID-19, malaria, dementia and tuberculosis. In addition, the top 3 AI technologies are ML, AI, and DL.

VOSviewer can show bibliometric mapping in three specific visualizations as depicted in Figs. 11, 12, and 13. Keywords had been labelled with coloured circles. The greatness of the circle is correlated with the appearance of key phrases in the titles and abstracts. The more frequently a keyword appears, the larger the size of the letters and circles. Therefore, the occurrence rate governed the size of letters and circles.

As said by [56], density views are beneficial for figuring out the general design of a map and considering the most critical parts. From Fig. 13, the research intuitively focuses on AI study. “Machine learning”, “artificial intelligence”, COVID-19, and “epidemiology” turn out to be necessary.

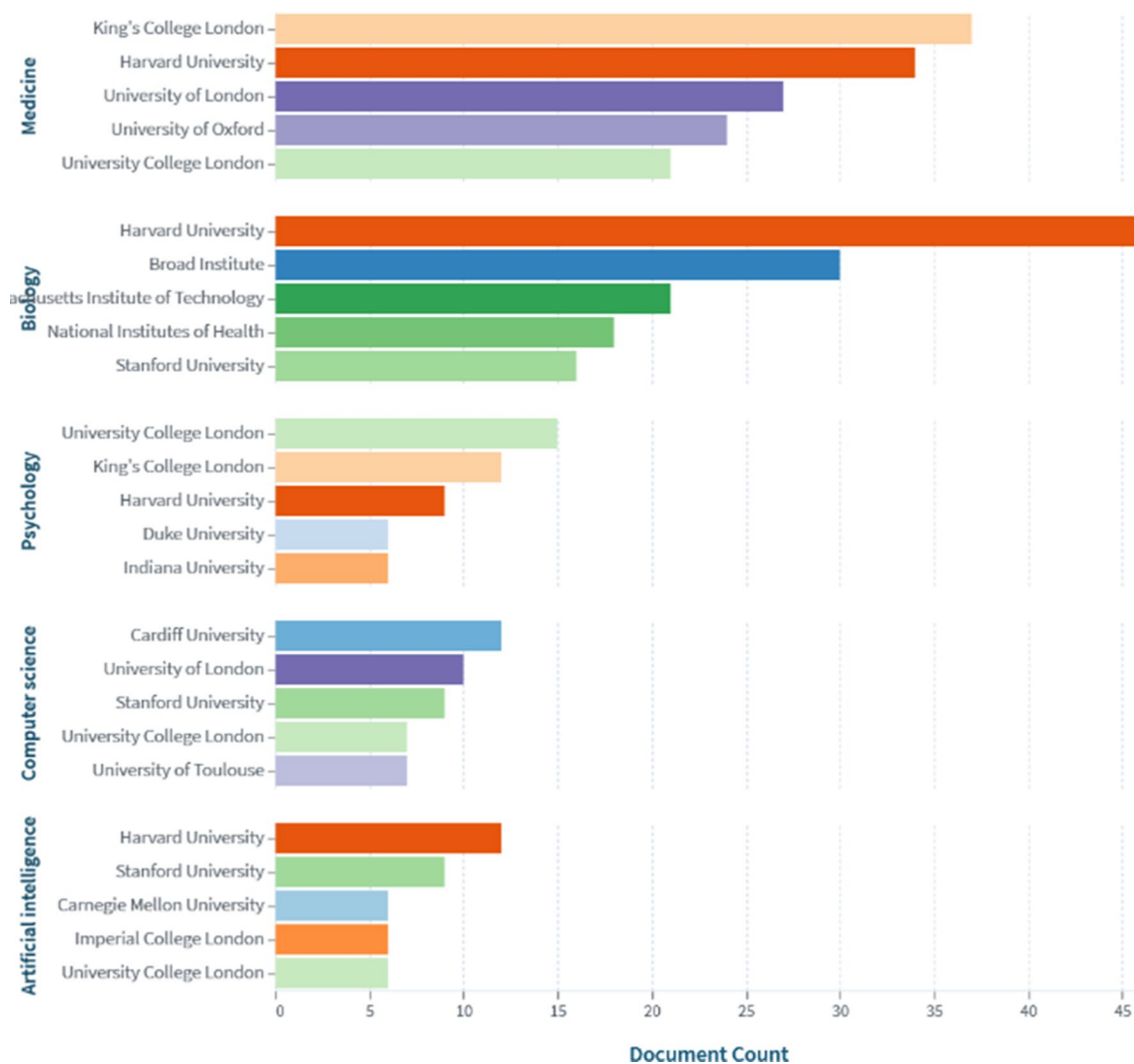


Fig. 9 Fields of study covered by the most active institutions

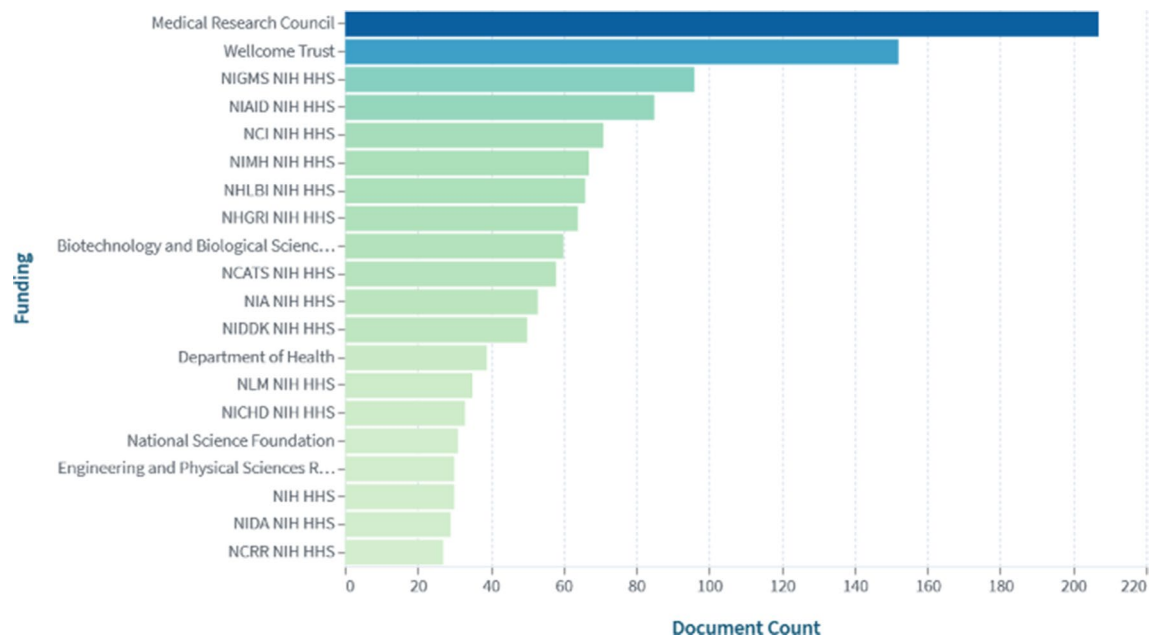
#### 4.13 Co-authorship analysis

After choosing five as the minimum number of authors' papers and deselecting the option to disregard documents with many authors, 153 of the 31,574 writers met the standards. Six different groupings or clusters were created by connecting all 150 authors together, as follows: according to Fig. 15, 38 authors were in cluster 1, 30 authors were in cluster 2, 12 authors were in cluster 3, 10 authors were in cluster 4, 9 authors were in cluster 5, and 7 authors were in cluster 6, showing that 106 authors have a close relationship and have contributed significantly to the field by cooperating. Also noteworthy is the fact that the study's normalization strategy was association.

Results in Table 5 indicate that Paul M. Thomson produced the most significant number of publications (i.e., 12 documents) with 1095 citations, in collaboration with

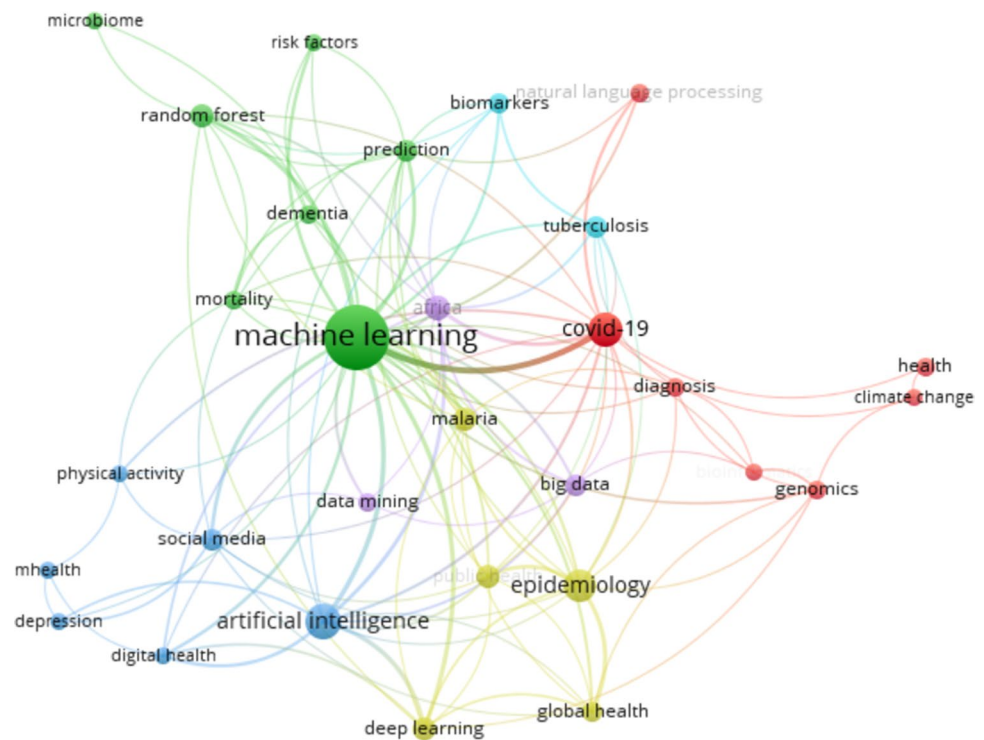
other authors, and the largest total link strength is 218. Dan J. Stein follows this with 20 publications and 899 citations with 208 as the total link strength, Dick J. Veltman with ten publications and 1004 citations with 200 as the total link strength, Neda Jahanshad with ten publications and 1066 citations with 197 as the total link strength, Henrik Walter with ten publications and 1045 citations with 193 as the total link strength. Thus, Paul M. Thomson is the most influential author in Africa's AI for healthcare research domain.

Researchers create learning networks, advance various modes of thought, and provide research problem solutions through this cooperative process. According to the number of their publications, Fig. 14 shows a network representation of highly prolific writers, with near frames denoting author partnerships on research. These frames are grouped into two, each representing a different research community. The number of co-authored publications between two researchers is



**Fig. 10** Funding bodies

**Fig. 11** Keywords co-occurrence network of AI-related publications



shown by the number of map lines defining their connection. The VOSviewer method was utilized to “visualize” writers with the least productivity of five documents.

The simplest display mode, the “network visualization” displayed in Fig. 14, illustrates the evolution of co-authorship and the emergence of some clusters. The second display

method, “Overlay visualization”, presented in Fig. 15, returns a network that brings the information of the chronological kind and iterations of co-authorship and cluster creation (see the legend in the bottom right corner of Fig. 15). Finally, the third display mode, “Density visualization”, shown in Fig. 16, clearly verifies the various densities of

**Table 4** Top keywords of the AI-related publications

| Keyword                 | Occurrences | Total link strength |
|-------------------------|-------------|---------------------|
| Machine learning        | 129         | 82                  |
| Artificial intelligence | 42          | 37                  |
| COVID-19                | 33          | 27                  |
| Epidemiology            | 35          | 27                  |
| Africa                  | 22          | 26                  |
| Deep learning           | 18          | 24                  |
| Random forest           | 18          | 17                  |
| Global health           | 14          | 16                  |
| Prediction              | 16          | 16                  |
| Public health           | 18          | 16                  |
| Digital health          | 11          | 14                  |
| Malaria                 | 19          | 14                  |
| Social media            | 16          | 14                  |
| Big data                | 16          | 10                  |
| Dementia                | 12          | 10                  |
| Tuberculosis            | 17          | 10                  |

the information provided in the network without displaying iterations and clusters like the other modes.

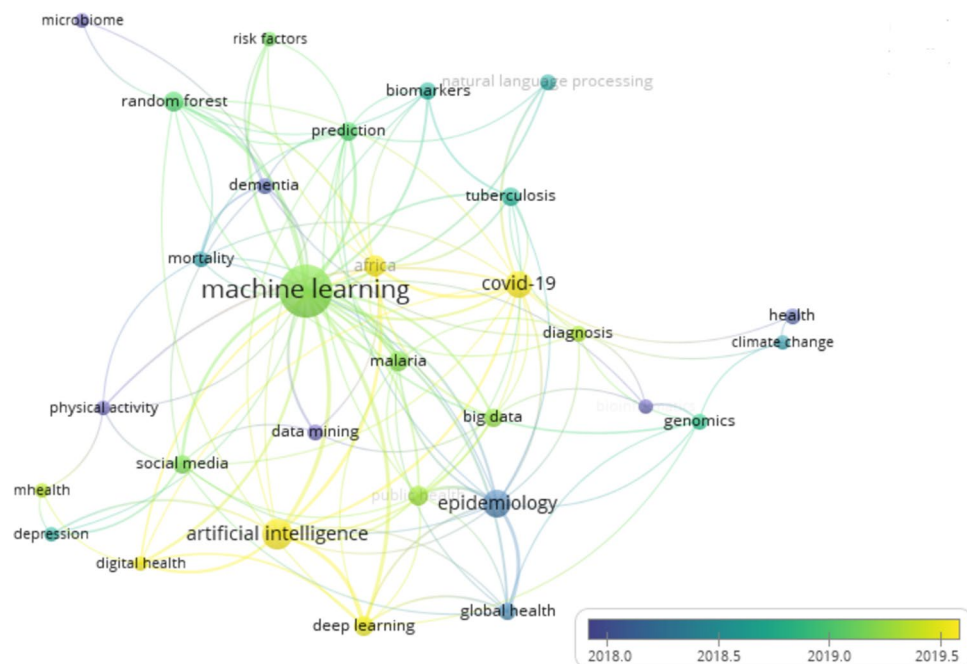
#### 4.14 Bibliographic coupling analysis

A similarity statistic between research sources called bibliographic coupling can be used to contrast various study topics [57]. Bibliographic coupling occurs when two documents frequently cite the third study. Each node in the

bibliographic coupling of journals is a journal, and each colour is a cluster. The outcome includes three groupings. PLOS One is one of the 19 things in Cluster 1, 8 items are in Cluster 2, and BMC genomics is one of the 7 items in Cluster 3. According to Table 6, the bibliometric coupling between the journals shows that PLOS One (TLS: 2385), Remote sensing (TLS: 1190), and Scientific reports (TLS: 952) have the strongest relationships. Figures 17, 18 and 19 display the network, overlay and density visualization maps for the bibliographic coupling of journals. The total link strength is 6225, and there were 383 links.

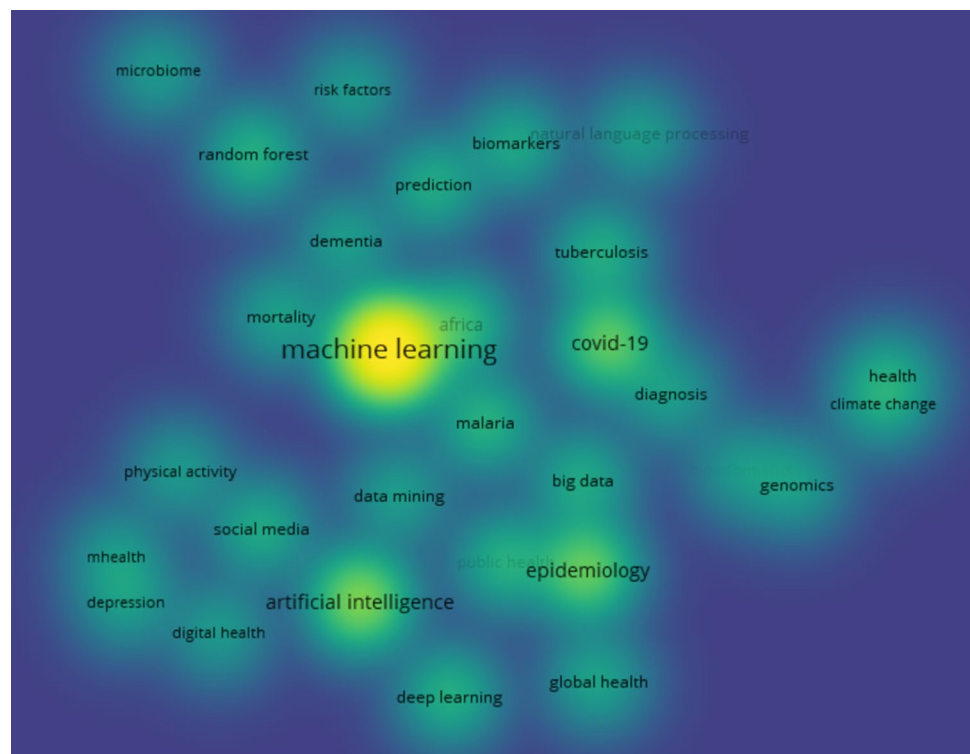
#### 4.15 Three plots field

“Biblioshiny”, an R-based package and the web-based interface of “Bibliometrix” were used to create the three plots field utilized in this study [44]. Figure 20 shows a three-field plot based on a Sankey diagram that shows the relationships between sources, authors, and keywords. By publishing in the sources on the left, the authors in the middle of the picture have developed the ideas on the right. Along with public health, prediction, artificial intelligence, and random forest, other important ideas such as epidemiology, dementia, deep learning, mortality, and diagnosis come to the fore. The authors who have written the most on these ideas are Stewart R, Li X, Cano J, and Wang Z. These authors frequently use Plos One, Journal of Biomedical Informatics, Nature Communications, and Plos Neglected Tropical Diseases while writing about these themes.

**Fig. 12** Overlay



**Fig. 13** Keywords density visualization map of AI-related publications



**Table 5** Most cited authors

| Author             | Documents | Citations | Total link strength |
|--------------------|-----------|-----------|---------------------|
| Paul M. Thompson   | 12        | 1095      | 218                 |
| Dan J. Stein       | 20        | 899       | 208                 |
| Dick J. Veltman    | 10        | 1004      | 200                 |
| Neda Jahanshad     | 10        | 1066      | 197                 |
| Henrik Walter      | 10        | 1045      | 193                 |
| Jair C. Soares     | 6         | 837       | 152                 |
| Lianne Schmaal     | 8         | 949       | 152                 |
| Sean N. Hatton     | 6         | 783       | 152                 |
| Andrew M. McIntosh | 5         | 768       | 148                 |
| Hans J. Grabe      | 5         | 768       | 148                 |
| Ian B. Hickie      | 5         | 768       | 148                 |
| Jim Lagopoulos     | 5         | 768       | 148                 |
| Katharina Wittfeld | 5         | 768       | 148                 |
| Ole A. Andreassen  | 7         | 939       | 144                 |
| Dara M. Cannon     | 6         | 938       | 143                 |
| Oliver Gruber      | 5         | 763       | 142                 |

#### 4.16 Trend topics

In the research on AI for healthcare, the authors' keywords are displayed according to the number of terms used most frequently. This research includes a section on topic trends, where Fig. 21 displays an overview of the topic's

evolution over time with a breakdown by year based on the authors' keywords. It displays writers' most recent keywords and terms they have used recently. Based on Fig. 21, the authors' keywords from 2014 to 2018 were "electronic health records." In 2016, the term "data mining" started to appear, and it has been used ever since. Since 2020, subjects with varying degrees of popularity have included "machine learning," "Africa," "artificial intelligence," "agriculture," "pandemic," and "COVID-19".

#### 4.17 Thematic map

Figure 22 illustrates the analysis of thematic maps into 4 theme quadrants based on density and centrality in this study. As a driving theme with high density and centrality, the upper right quadrant needs to be developed and is crucial to be researched further. However, there were no themes identified in this quadrant. In addition, the upper left quadrant exhibits a particular and uncommon motif while exhibiting significant development, as shown by high density but low centrality. Big data are the central idea in this sector.

In addition, themes that have been around for a while but have had a decreasing trend and low centrality can be found in the lower left quadrant, but no motifs were found there. Finally, a fundamental motif with high centrality but low density can be seen in the lower right quadrant. Machine learning, artificial intelligence, epidemiology, COVID-19, and Africa are the themes in this quadrant.

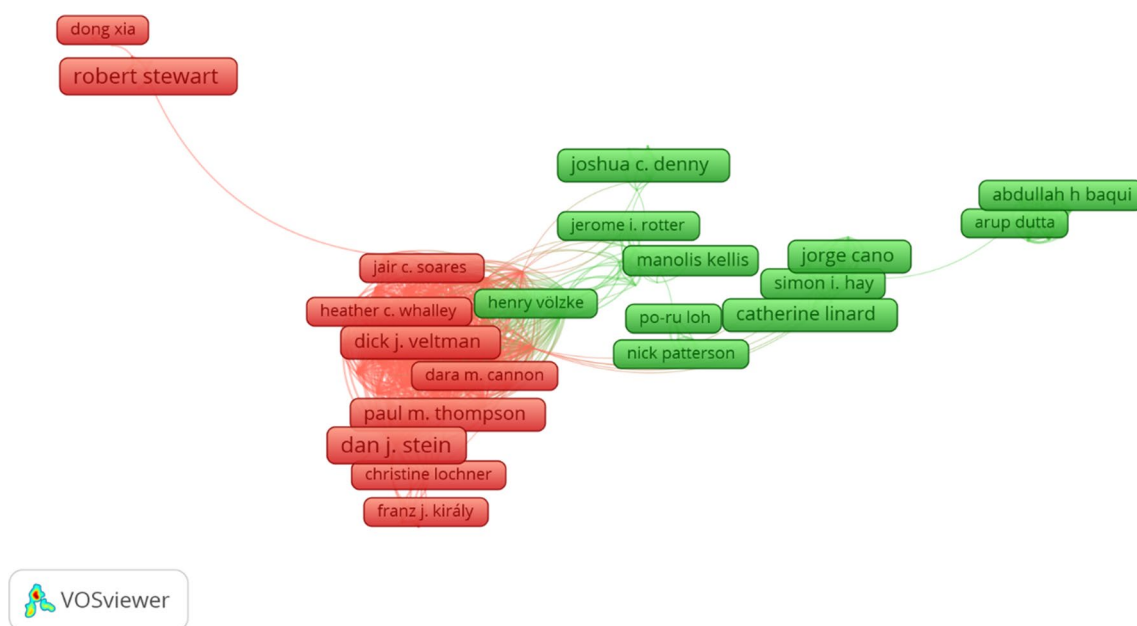


Fig. 14 Network visualization of highly productive authors

## 5 Discussion

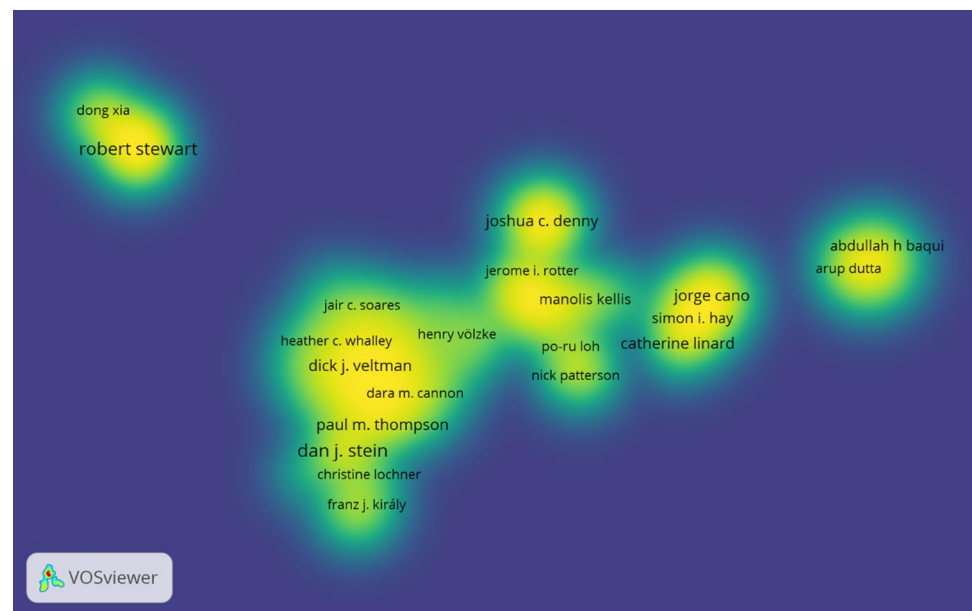
This study explored some stimulating results regarding AI for healthcare-related publications in Africa. It offers a thorough review of the AI-related research undertaken in the healthcare industry. Furthermore, a series of figures that may be utilized to grasp the textual content better

were produced using VOSviewer and bibliometrix software. In addition, several tables were created to tabulate the data. It was determined that publishing growth started slowly during the first 40 years and picked up speed in 2000. As a result, it is projected that the research trends in AI for healthcare will keep growing in the years to come as more people spend their days using electronic devices.



Fig. 15 Overlay visualization display mode

**Fig. 16** Density visualization display mode



In addition, it was discovered that organizations from the United Kingdom, the United States, and South Africa had the most significant number of publications, accounting for more than half of them. These prominent AI research participants can serve as potential international partners and an example for other nations with fewer publications to follow. The United Kingdom has seven institutes ranked in the top 10 concerning the number of AI for healthcare-related publications. Moreover, Harvard University is the most active institution, with the maximum number of publications. The journal, *Annals of Tropical Medicine and Public Health*, ranked at the top regarding the AI for healthcare-related scholarly works.

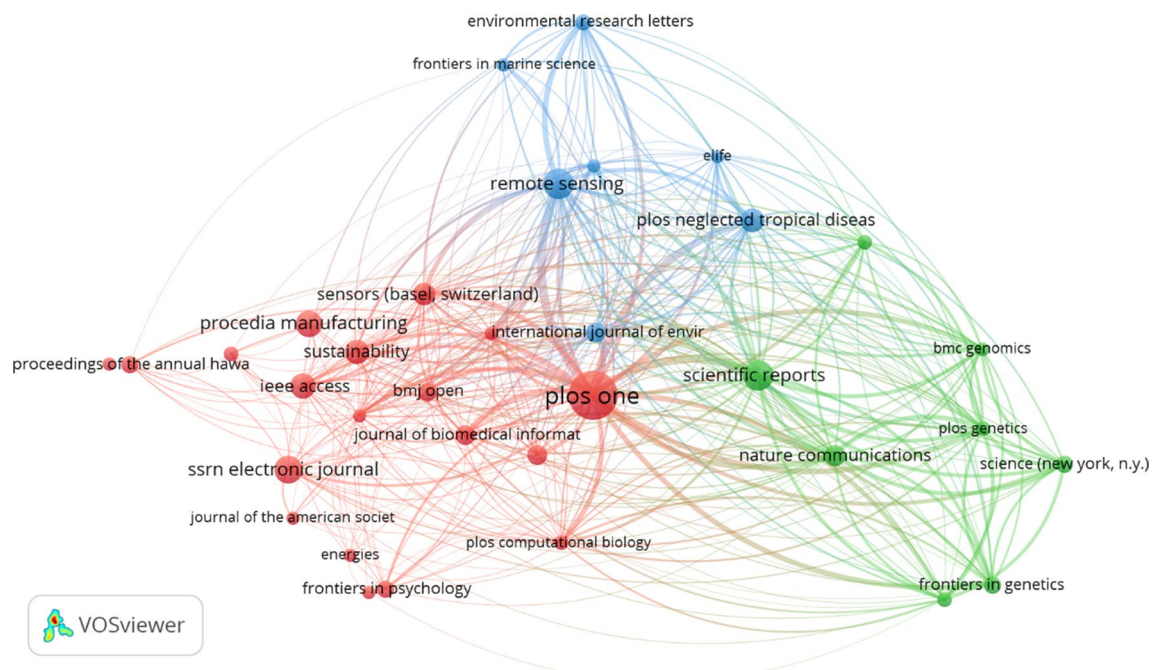
In collaboration with other authors, Paul M. Thomson was the most influential author, with 12 publications and 1095 citations. The largest total link strength is 218. Dan J. Stein followed him with 20 publications and 899 citations, with 208 as the total link strength. Furthermore, Elsevier is

the first on the list of top publishers in the world with 620 publications and Harvard University is at the top in the fields of AI and Biology. Thematic map of the keyword shows that machine learning, artificial intelligence, epidemiology, COVID-19, and Africa are in the lower right quadrant (Basic theme). Since 2020, subjects with varied degrees of popularity have included “machine learning”, “Africa”, “artificial intelligence”, “agriculture”, “pandemic”, and “COVID-19”. The findings of this study also show that the United Kingdom is the most productive and cooperative nation and that African scholars are significantly under-represented in these publications.

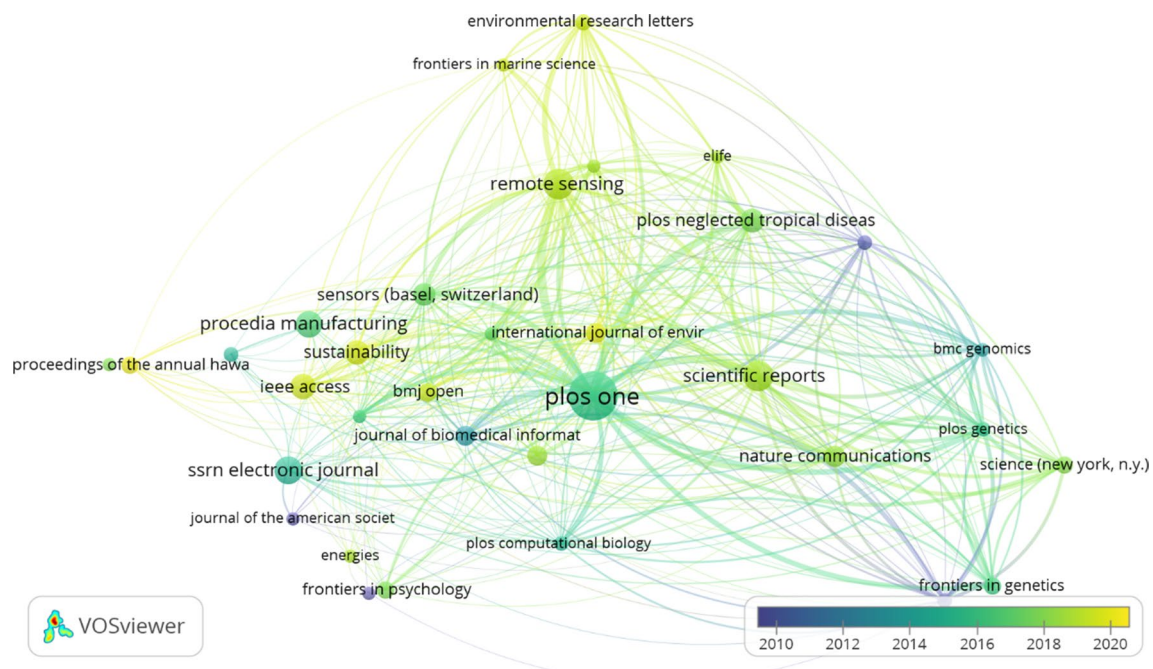
The need to undertake this work was driven by several critical factors. Initially, as AI increasingly plays a critical role in global healthcare, it is essential to assess Africa’s position in adopting this technology. Understanding the current state of artificial intelligence research in African healthcare is essential for developing AI solutions

**Table 6** Top cited journals

| Source                             | Documents | Citations | Total link strength |
|------------------------------------|-----------|-----------|---------------------|
| Plos one                           | 124       | 5639      | 2385                |
| Remote sensing                     | 49        | 1000      | 1190                |
| Scientific reports                 | 48        | 1394      | 952                 |
| Plos genetics                      | 12        | 1087      | 696                 |
| Plos neglected tropical diseases   | 29        | 673       | 640                 |
| Nature communications              | 24        | 2348      | 608                 |
| BMC genomics                       | 10        | 564       | 488                 |
| Frontiers in genetics              | 17        | 388       | 470                 |
| Science (New York, n. y.)          | 15        | 4869      | 459                 |
| American journal of human genetics | 11        | 5191      | 452                 |



**Fig. 17** Network visualization

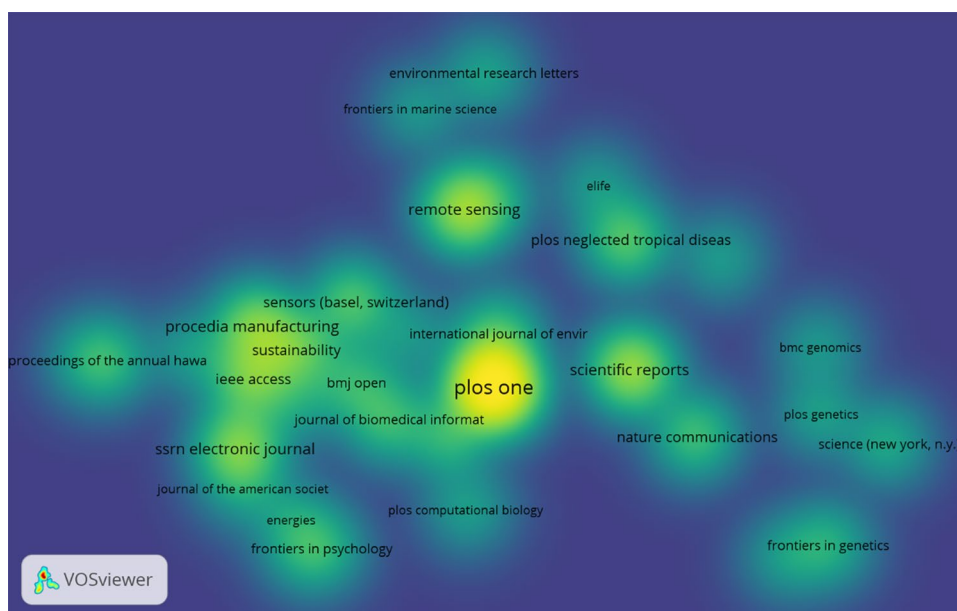
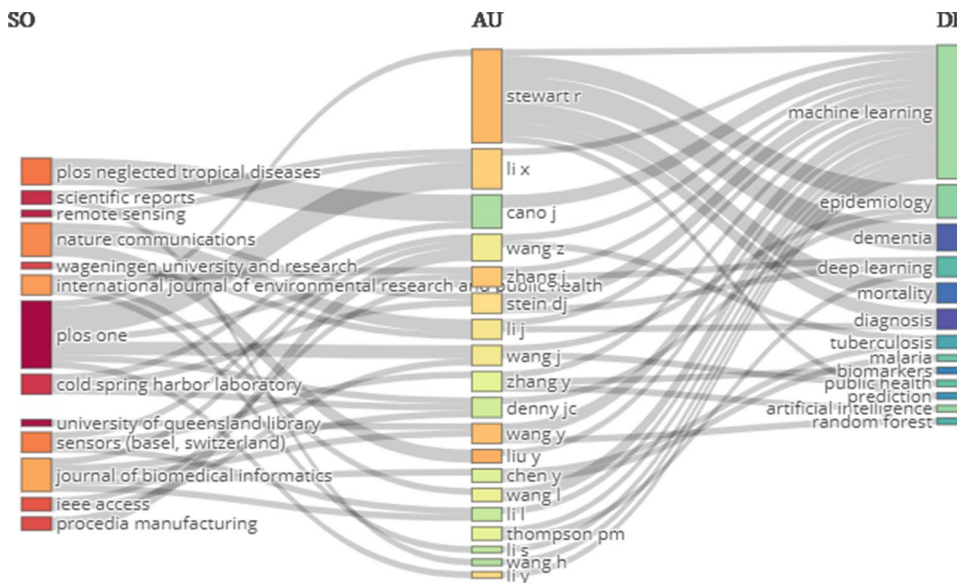


**Fig. 18** Overlay visualization

that are tailored to the specific conditions that exist on the continent, which include the prevalence of infectious diseases and the scarcity of resources. In addition, this study attempted to identify gaps in research, areas with high activity, and possible collaborations by conducting

a bibliometric analysis. The results of this study would provide significant insights for researchers, policymakers, and healthcare practitioners working toward the improvement of healthcare outcomes in Africa.



**Fig. 19** Density visualization**Fig. 20** Sources (left), authors (middle), keywords (right)

Moreover, this work sought to shed light on the thematic areas of AI research in African healthcare. Africa's healthcare landscape is diverse, with varying disease burdens and infrastructural constraints across regions. Therefore, it was essential to identify which healthcare domains were receiving the most attention within AI research. By doing so, this analysis could help prioritize research efforts and allocate resources more effectively to address pressing healthcare challenges, whether related to diagnostics, treatment, or health system optimization [58]. In essence, this study aimed to bridge the gap between AI capabilities and healthcare needs in Africa, ultimately contributing to the development of contextually relevant and impactful AI applications in the field of healthcare on the continent.

While Africa has faced historical challenges in terms of research infrastructure, funding, and access to advanced technology, it would be simplistic to conclude that AI research in Africa is universally lagging behind. However, Africa is a diverse continent with significant variations among countries and institutions. South Africa, Morocco, Uganda, Egypt, Mauritius, Ghana, and Rwanda are just a few of the African countries that have made significant advancements in artificial intelligence (AI) research and application, and there are an increasing number of African institutions and research centres that are actively contributing to the subject. In addition, artificial intelligence has the potential to present exceptional prospects in Africa, such as the improvement of healthcare and agricultural practices.



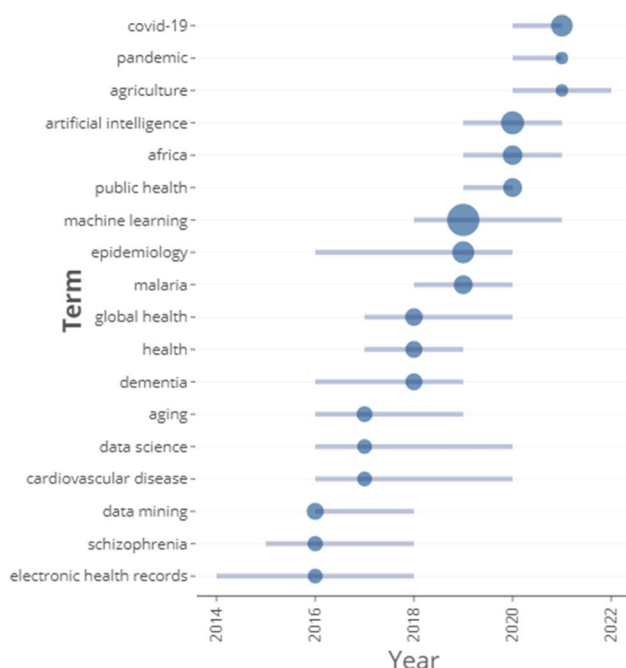


Fig. 21 Trend topics

In spite of the fact that inequities do exist, it is essential to acknowledge and encourage the growing hubs of brilliance and innovation that can be found all over the African continent.

AI research in healthcare is a global effort, and different regions have their own strengths, challenges, and research priorities [59]. While the current study provides valuable

insights into the status of AI research in African healthcare, it's crucial to acknowledge that each region faces unique healthcare issues and have different levels of AI research infrastructure. Moreover, the volume and focus of research can vary significantly. Therefore, it's more constructive to view AI research in healthcare as a collaborative global endeavour, where knowledge sharing and cross-regional partnerships can lead to advancements that benefit healthcare systems worldwide. Hence, efforts should be focused on building research capacity, fostering collaborations, and providing resources to empower AI research in Africa, ultimately dispelling the assumption of a significant research gap and contributing to the global AI community.

The study's emphasis on AI research involving Africa highlights a pivotal ethical consideration concerning the equitable and unbiased performance of AI models. As evidenced in the literature, biases can inadvertently infiltrate AI algorithms, resulting in disparities in healthcare outcomes. Consequently, there is a compelling rationale for further research that encompasses representative samples from diverse ethnic backgrounds, with a particular focus on Africans, within the realm of AI for healthcare advancement. However, ensuring fairness and ethical usage of AI in healthcare transcends mere technical challenges; it is a matter of social justice. Inclusive AI research spanning a wide range of populations is indispensable for mitigating biases and disparities in healthcare outcomes. This paper underscores the ongoing necessity for research endeavours aimed at addressing this critical ethical concern and fostering equitable AI-driven healthcare not only in Africa but also on a global scale.

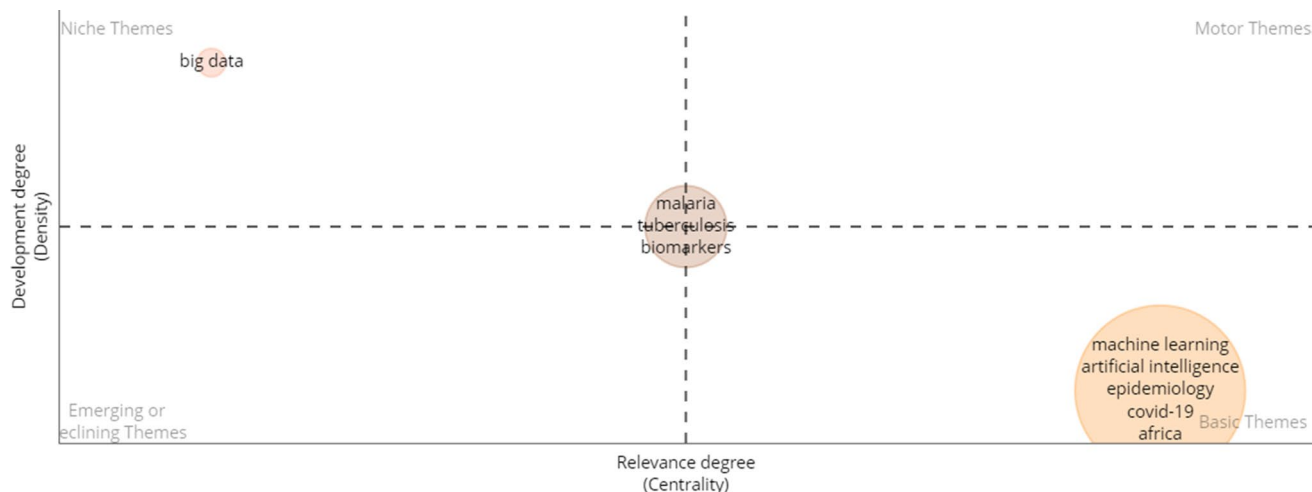


Fig. 22 Thematic map

## 6 Conclusions

It should be noted that, AI research in Africa can empower local researchers and institutions to develop AI models and solutions tailored to the continent's healthcare landscape. This includes addressing diseases that are more prevalent in Africa, such as neglected tropical diseases. Furthermore, AI research can enhance the capacity of African healthcare systems to manage health data, which is crucial for evidence-based policymaking and improving health outcomes.

Moreover, ethical considerations and data privacy concerns play a vital role in AI implementation [60]. African countries, such as others, must grapple with questions of patient data security and privacy, regulatory frameworks, and ethical guidelines, all of which need to be addressed before widespread AI adoption [61]. To bridge the implementation gap, African nations need targeted strategies that account for their unique challenges and strengths. Collaborations with international partners, investment in local capacity building, and the development of AI solutions tailored to the African context are all essential steps toward leveraging AI's potential to improve healthcare outcomes across the continent.

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**Data availability** The data that support the findings of this study are available from the corresponding author, [TSK], upon reasonable request.

## Declarations

**Conflict of interest** On behalf of all authors, the corresponding author states that there is no conflict of interest.

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